DK1 imbalance: methods

github.com/jacobazu/dk1-imbalance-forecasting · October 2026

Forecasts of the chance that DK1 is regulated up, down or not at all in each quarter-hour, and of the spread between the imbalance price and spot. Every hour for the next four hours, and once a day for tomorrow.

Inputs	Calendar, the spot price, wind and solar forecasts made 48 hours ahead, recent imbalances, and real-time grid data for the next hour. Before the auction the price forecast stands in for spot.
Models	Multinomial lasso (Tibshirani, 1996; Friedman et al., 2010), XGBoost (Chen and Guestrin, 2016) and their average, refitted weekly. Benchmark: how often each direction happened over the last 28 days. Spread: past spreads at a similar spot price, weighted by the direction probabilities.
Scoring	Log loss and pinball loss against the benchmark, Diebold–Mariano tests with a Newey–West variance (Diebold and Mariano, 1995; Newey and West, 1987), calibration, and the value of picking a side (Bottieau et al., 2020).
Day-ahead to imbalance	A test of whether tomorrow's spread can be forecast before the auction: long or short 1 MW per quarter on the day-ahead auction, settled at the imbalance price, after fees. Compared with always long, always short and the benchmark with Newey–West and Holm. Paper only.
Rules	Each forecast only uses data published when it is made, and the backtest uses the same inputs. Test rules are written before the test.

References

Bottieau, J. et al. (2020). Very-short-term probabilistic forecasting for a risk-aware participation in the single price imbalance settlement. *IEEE Transactions on Power Systems*.

Chen, T. and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *KDD*.

Diebold, F. X. and Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*.

Friedman, J., Hastie, T. and Tibshirani, R. (2010). Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*.

Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*.

Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *JRSS B*.